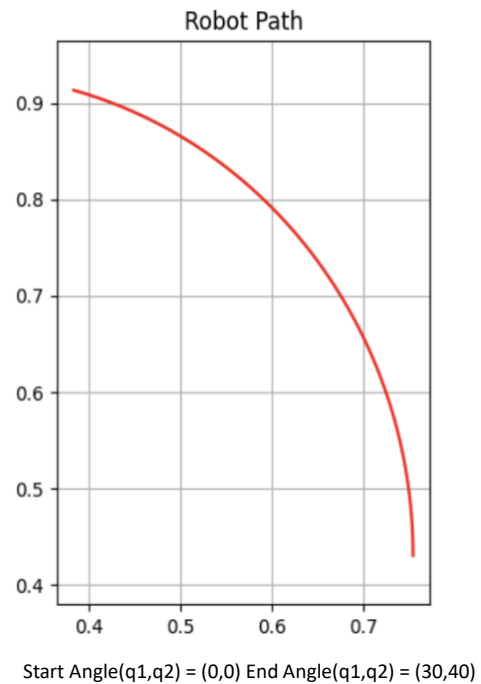
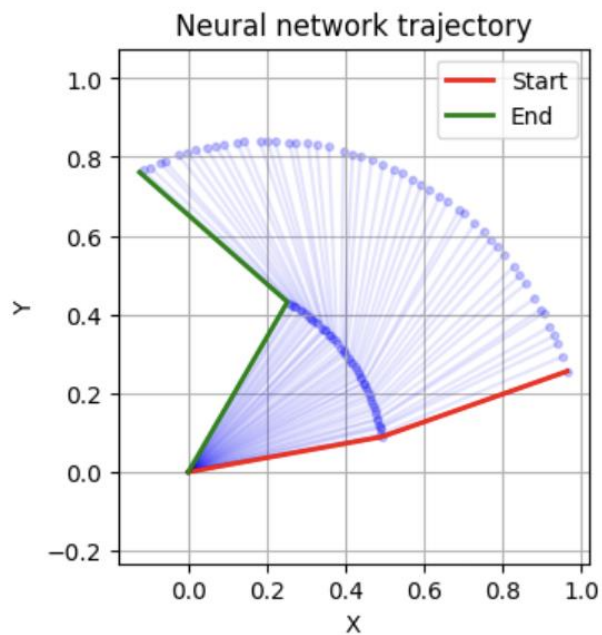




The faint blue lines represent the intermediate states, showing a smooth, continuous interpolation of joint angles. In the first graph, we can see both links of the arm moving, which illustrates how the neural network is solving the Inverse Kinematics problem by adjusting two different angles simultaneously. The neural network basically learns to invert these equations through training.

In Plot 2 created by optimization, by using a numerical solver like minimize, we can specifically target a cost function such as squared joint acceleration to ensure the smoothest possible transition. Optimization includes the constraints of the joints ensuring movement of robot well within the limits.